Problem Set: MLE, GMM, and ARMA

Bryan Kelly
Yale University
bryan.kelly@yale.edu

Guidelines

No hand-written answers. Perform estimation from scratch using your own code (i.e., do not use built-in regression, GMM, or MLE functions, though you may use a built-in numerical optimizer and built-in variance-covariance functions). Include your code with your solutions.

Questions

1. Pull JKP factor data from https://jkpfactors.com/data
Select **four factors** of your choice and briefly justify why you find them interesting. Use data from 1975–2025.

- Plot and test the first 12 autocorrelations and partial autocorrelations. Based on this, state in *one sentence* which ARMA model you expect to fit best for each factor and why.

- Estimate AR(1), ARMA(1,1), and your ARMA(p,q) model of choice for each factor via:

  (a) Regression

  (b) Exact MLE assuming a Gaussian model (write down the exact likelihood)

- Split your sample into 1975–2015 (estimation) and 2016–2025 (evaluation). Re-estimate your models on the training sample and compare their out-of-sample forecast. Does the model you selected based on ACF/PACF perform best out of sample?

- For **one factor of your choice**, write a one-paragraph explaining what your preferred model implies for next month's expected return on that factor.

2. **PhD*** The data set `wsj-counts.csv` contains occurrence frequency of various phrases from *The Wall Street Journal*. Let $c_{i,t}$ denote the counts of phrase i at time t. Consider an autoregressive dynamic model for expected phrase counts:

$$E_{t-1}[c_{i,t}] = \delta_0 + \delta_1 c_{i,t-1}$$

Tasks:

- Estimate this model via:

(a) OLS regression

(b) Exact MLE assuming a Poisson model for each series (hint: you must represent the conditional expectation within the Poisson distribution). Write down the exact likelihood.

(c) Describe and explain differences in your results from regression versus MLE.

- Test estimated ARMA parameters using OLS, ML-based Wald test, likelihood ratio test, and Lagrange multiplier test. Explain how you set up each of these tests.

3. PhD* For each phrase i in `wsj-counts.csv`, estimate the Poisson parameter λ_i via GMM using the Poisson model mean and variance as the moment conditions.

- Write down your formulation of the GMM problem.
- Describe and explain differences in your results from GMM versus MLE.
- Perform an asymptotic test of the overidentifying restriction. Interpret your results.